GCSE PHYSICS · TEACHING DECK

Energy Stores & Changes

Deck 3 of 4 — Elastic potential energy & bringing it together

AQA 8463 | Edexcel 1PH0 | OCR J249 / J250

Mirrors kinetic energy — then links every equation so far into one falling-object idea



Where we're going — 4 decks



1

Systems, transfers & how to describe them

All tiers

2

Kinetic & gravitational potential energy

All tiers

YOU ARE HERE

3

Elastic potential energy & bringing it together

Higher stretch

4

Quick recap & multi-step challenges

Review

All tiers again — but the second chunk carries a Higher stretch flag: rearranging m g h = ½ m v² to v = √(2 g h).

STARTER *~1 min*



"As you stretch a spring, what fills up? And what happens the instant you let go?"

Push toward: The elastic potential store fills as you stretch it; releasing empties it into the kinetic store.

Tutor: *Links storing energy to releasing it — sets up the equation as the natural next step.*

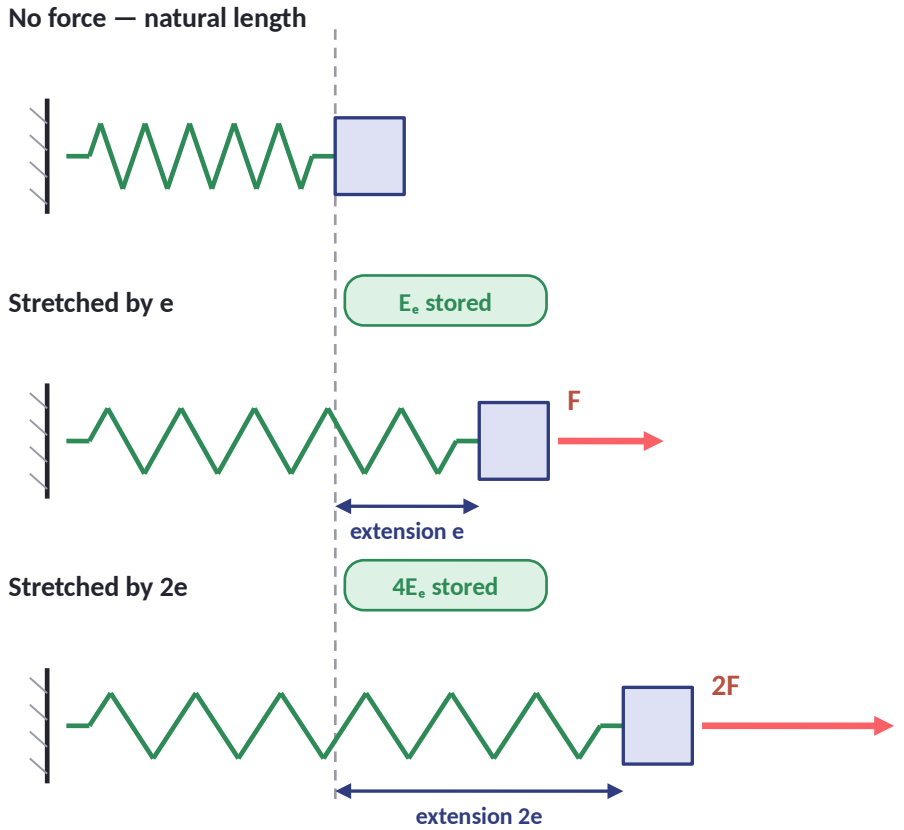
THEIR IDEAS

Elastic potential energy — $E_e = ½ k e^2$

- Elastic potential energy is the energy stored in a stretched or squashed spring (or other elastic object).
- $E_e = ½ k e^2$ (EPE in J = 0.5 × spring constant k in N/m × (extension e in m)^2), provided the limit of proportionality is not exceeded. Edexcel writes $½ k x^2$.
- Extension is squared, so — like kinetic energy — doubling the stretch quadruples the stored energy.

$E_e = ½ k e^2$

J = 0.5 × N/m × m^2 · valid only up to the limit of proportionality



extension = stretched length − natural length. Double e → four times E^e.

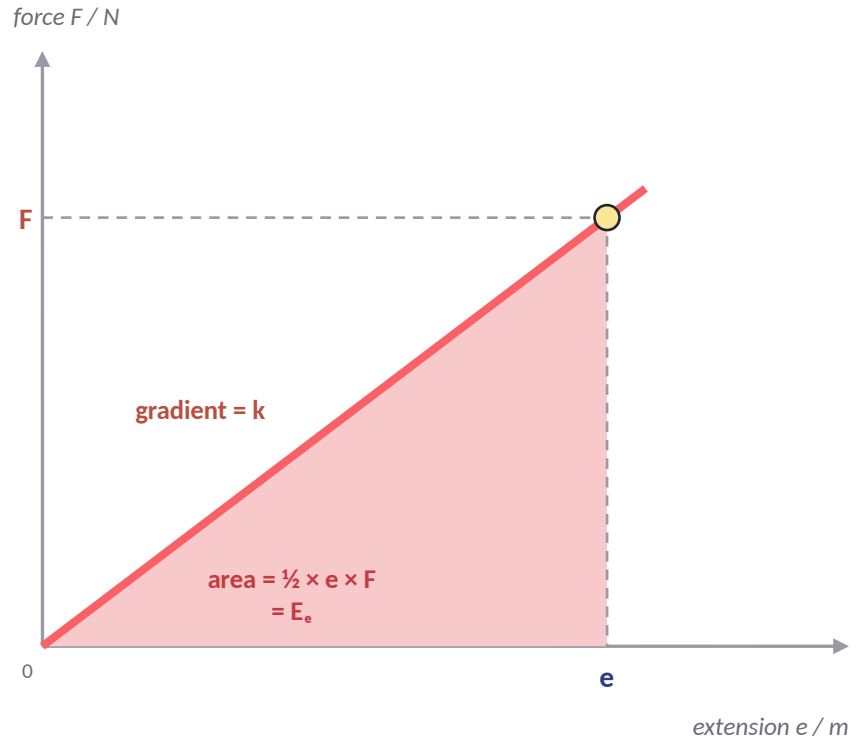
Where marks are lost: missing the condition ("provided the limit of proportionality is not exceeded"); squaring k instead of e; using total length instead of extension.

Force, extension and the energy stored

- Extension e = stretched length − natural length. Convert cm to m before substituting.
- Up to the limit of proportionality, F = k e: the graph of force against extension is a straight line through the origin, and its gradient is the spring constant k (N/m).
- **The area under the line is the energy stored: ½ × e × F = ½ × e × (k e) = ½ k e².**
- Rearranged: $k = 2E_e \div e^2$ and $e = \sqrt{(2E_e \div k)}$.

$$k = 2E_e \div e^2$$

N/m = J ÷ m² · rearranged for extension: $e = \sqrt{(2E_e \div k)}$



Straight line = F = k e (up to the limit of proportionality). Area under it = energy stored.

Where marks are lost: forgetting to square e when finding k; using the whole spring length; dropping the 2 in the rearrangement.

WORKED EXAMPLE

Tutor models the method — answer comes last

A spring (k = 200 N/m) is stretched 0.10 m. EPE stored?

METHOD

- Write the equation: $E_e = ½ k e^2$
- Substitute: $E_e = ½ \times 200 \times 0.10^2$
- Square the extension first: $0.10^2 = 0.01$, then multiply through.

WORK IT THROUGH

Answer: $E_e = ½ \times 200 \times 0.01 = 1.0$ J.

Mark scheme: *1 for equation · 1 for substitution (squared extension) · 1 for 1.0 J.*

YOUR TURN — PART 1

Elastic potential energy ($E_e = \frac{1}{2} k e^2$) — answers on the next slide.

1 State the EPE equation with units and the condition for using it.

2 A spring (k = 50 N/m) is compressed 0.20 m. Energy stored?

3 A spring stores 0.9 J when stretched 0.30 m. Find its spring constant.

4 If the extension of a spring doubles, what happens to the stored energy?

Working space is squared on purpose — students calculate, tutor annotates.

ANSWERS — PART 1

Tutor reveal

1

$E_e = \frac{1}{2} k e^2$ (J, N/m, m), valid up to the limit of proportionality.

2

$\frac{1}{2} \times 50 \times 0.20^2 = 25 \times 0.04 = 1.0$ J.

3

$k = 2E \div e^2 = 1.8 \div 0.09 = 20$ N/m.

4

It becomes four times bigger (×4).

STARTER *~1 min*



"A ball dropped from a height speeds up as it falls. Which store empties, which fills — and if we ignore air resistance, are the two amounts equal?"

Push toward: GPE store empties as the KE store fills — with no air resistance, the amounts are exactly equal.

Tutor: *Bridges directly from Deck 2's GPE equation into today's big idea: conservation across a falling object.*

THEIR IDEAS

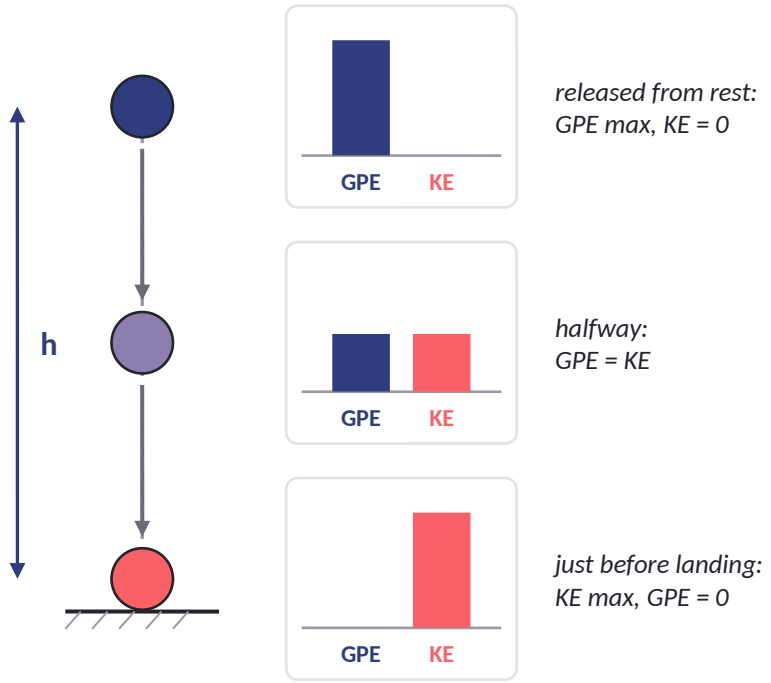
Conservation of energy: GPE lost = KE gained

HIGHER STRETCH — rearranging m g h = ½ m v² to v = √(2 g h) is where Higher students are pushed.

- When a system changes, energy shifts store to store. Ignoring friction and air resistance, energy is conserved: GPE lost = KE gained.
- **That links the two equations: m g h = ½ m v². The mass cancels, giving v = √(2 g h) — handy for "how fast does it land?"**
- In reality some energy is dissipated — transferred by heating to the thermal store of the surroundings by friction and air resistance.

$$m\,g\,h = \tfrac{1}{2}\,m\,v^2 \;\rightarrow\; v = \sqrt{(2\,g\,h)}$$

Mass cancels from both sides — this only works because both equations contain m

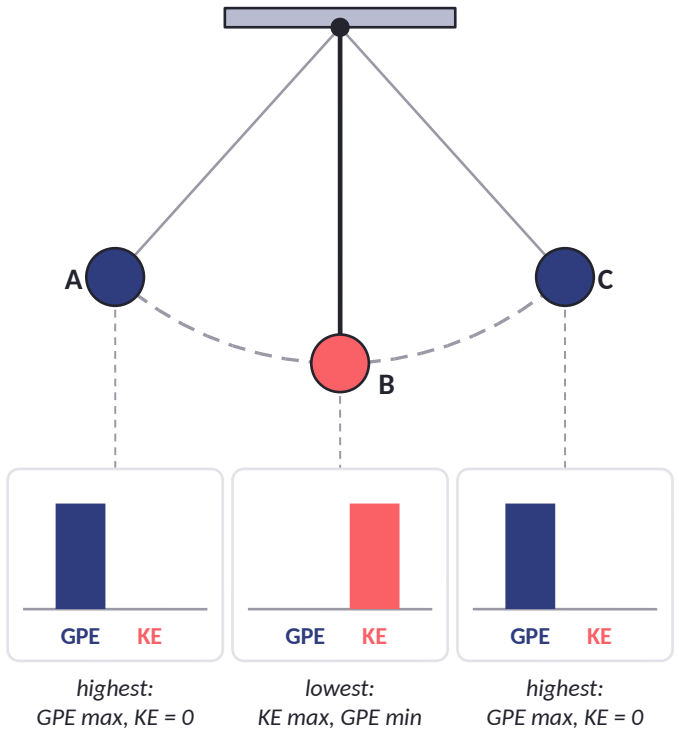


GPE lost = KE gained (no air resistance)

Where marks are lost: forgetting to state "assuming no air resistance"; not cancelling mass and getting stuck; treating dissipated energy as destroyed rather than transferred.

Swings and bounces: energy moving between stores

- A pendulum: at the highest points the gravitational potential store is full and the kinetic store is empty; at the lowest point it is the other way round.
- Total energy stays the same (ignoring dissipation) — the bars just change shape: GPE + KE = constant.
- Real pendulums slow down: air resistance and friction at the pivot dissipate energy by heating to the thermal store of the surroundings, so each swing is a little lower.

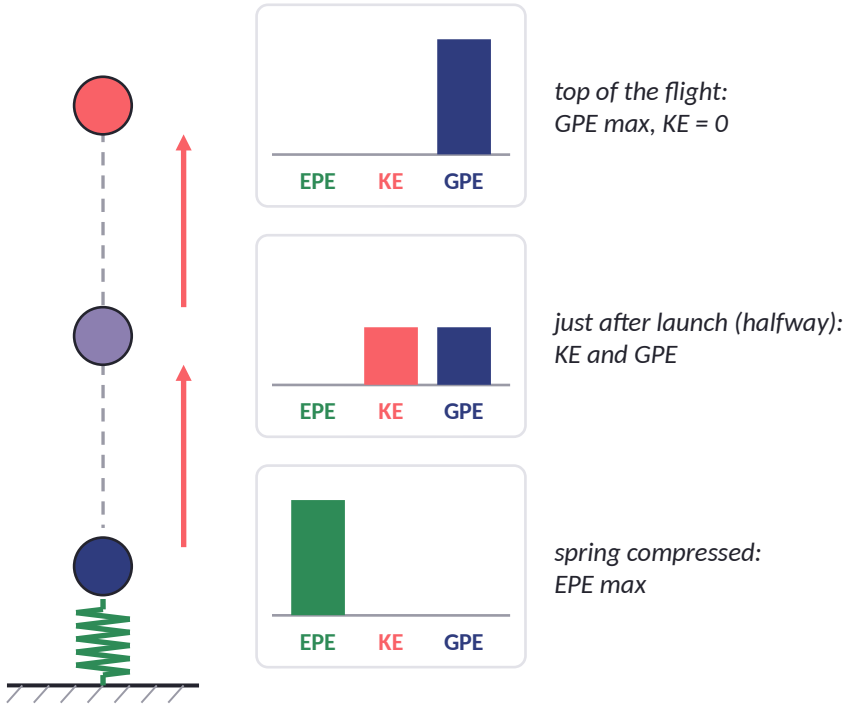


each swing a little energy is dissipated to the thermal store of the air and pivot, so the swing dies away

Where marks are lost: saying energy is "lost"; swapping which store is full at the top and the bottom; forgetting the thermal store of the surroundings.

Chaining the stores: spring → kinetic → gravitational

- A spring launches a ball upwards: elastic potential store empties → kinetic store → gravitational potential store at the top.
- **With no losses, set the energies equal: ½ k e² = m g h (everything the spring stored ends up as GPE).**
- Method: (1) find E_e from ½ k e²; (2) write E_e = m g h; (3) rearrange h = E_e ÷ (m g). Keep the intermediate energy in joules.



½ k e² = ½ m v² = m g h (no losses)

Where marks are lost: skipping the intermediate energy; using cm instead of m for the compression; forgetting that some energy is dissipated in a real launch, so the real height is less.

WORKED EXAMPLE

Tutor models the method — answer comes last

A 2 kg ball is dropped from 5 m (g = 9.8, no air resistance). Find its speed as it lands.

METHOD

- GPE lost = m g h = 2 × 9.8 × 5 = 98 J.
- By conservation of energy, this all becomes KE: ½ m v² = 98.
- Rearrange for v: v = √(2 × 98 ÷ 2).

WORK IT THROUGH

Answer: v = √98 ≈ 9.9 m/s. (Or straight from v = √(2 g h) = √(2 × 9.8 × 5) ≈ 9.9 m/s.)

Mark scheme: *1 for GPE = 98 J · 1 for equating to ½mv² · 1 for v ≈ 9.9 m/s.*

YOUR TURN — PART 2

Bringing the stores together — answers on the next slide.

1 State what “energy is conserved” means for a closed system.

2 A 0.5 kg ball falls 1.8 m (g = 9.8, no air resistance). Find its landing speed.

3 When air resistance is not ignored, where does the “lost” energy go?

4 A pendulum swings. Describe how the stores change from the highest point to the lowest.

Working space is squared on purpose — students calculate, tutor annotates.

ANSWERS — PART 2

Tutor reveal

1

The total energy stays the same — it only moves between stores, none is created or destroyed.

2

$v = \sqrt{(2 \times 9.8 \times 1.8)} = \sqrt{35.28} \approx 5.9$ m/s.

3

Transferred by heating to the thermal store of the air and surroundings (dissipated).

4

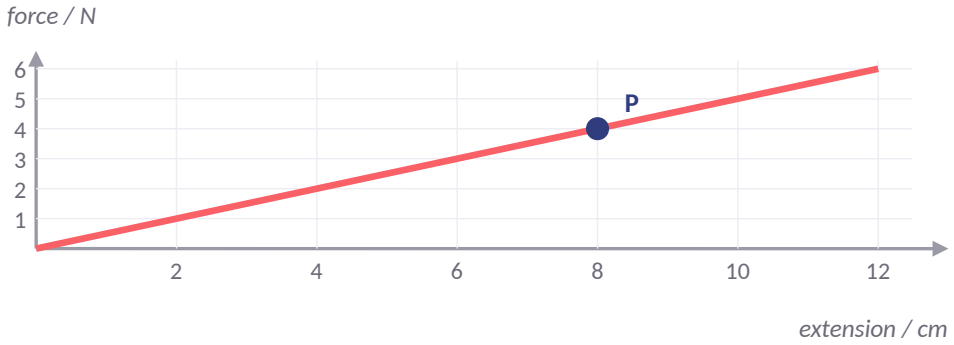
At the top: full GPE, zero KE; falling: GPE → KE; at the bottom: maximum KE, minimum GPE — then it reverses.

EXAM PRACTICE — SET A

Elastic potential energy · 15 marks in total · marks shown in [brackets] · mark scheme on the next slide

Q1 *Graph · calculation* **[7 marks]**

A student hangs masses from a spring. The spring obeys Hooke's law over the whole range shown.



(a) Use point P to calculate the spring constant in N/m. [3]

(b) Calculate the energy stored in the spring at point P. [2]

(c) The extension is doubled. State what happens to the stored energy and explain why. [2]

Q2 *Chained calculation · explain* **[8 marks]**

A toy launcher uses a spring (spring constant 600 N/m) to fire a ball of mass 12 g straight up. The spring is compressed by 4.0 cm. g = 10 N/kg.

(a) Calculate the elastic potential energy stored in the compressed spring. [3]

(b) All of this energy is transferred to the ball's gravitational potential store. Calculate the maximum height reached. [3]

(c) The ball actually rises less high than this. Explain why. [2]

MARK SCHEME — SET A

Tutor reveal · M = method, A = accuracy, B = independent mark, ECF = error carried forward

Q1

[7 marks]

B1 (a) reads F = 4.0 N at e = 8.0 cm and converts e = 0.080 m

M1 k = F ÷ e = 4.0 ÷ 0.080

A1 50 N/m (any valid point on the line)

M1 (b) $E_e = ½ k e^2 = ½ \times 50 \times 0.080^2$ (ECF k)

A1 0.16 J

B1 (c) four times bigger (×4)

B1 because E_e is proportional to e^2 (extension is squared)

Unit marks: k in N/m, E_e in J. Using 8.0 (cm) unconverted gives 6.25 N/cm: max 2.

Answer: (a) 50 N/m (b) 0.16 J

Q2

[8 marks]

B1 (a) 4.0 cm = 0.040 m

M1 $E_e = ½ k e^2 = ½ \times 600 \times 0.040^2$

A1 0.48 J

B1 (b) 12 g = 0.012 kg

M1 h = E_e ÷ (m g) = 0.48 ÷ (0.012 × 10) (ECF from (a))

A1 4.0 m

B1 (c) air resistance acts on the ball (accept friction with the launcher)

B1 so some energy is dissipated / transferred by heating to the thermal store of the air and surroundings, leaving less to transfer to the gravitational potential store

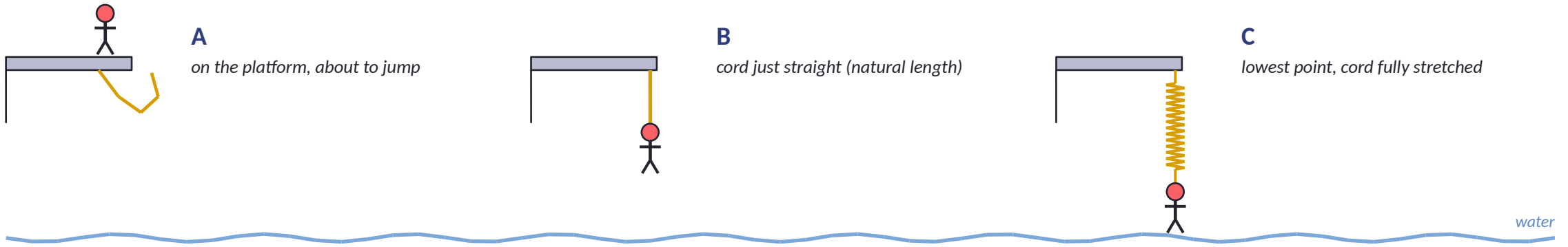
Answer: (a) 0.48 J (b) 4.0 m

EXAM PRACTICE — 6-MARK QUESTION

Levels-based · 6 marks in total · marks shown in [brackets] · mark scheme on the next slide

Q3 *Extended response · levels* **[6 marks]**

*A bungee jumper steps off a platform. The jumper falls, the elastic cord becomes taut and stretches, and the jumper reaches the lowest point C before being pulled back up. Describe the energy transfers between A, B and C, and explain why the jumper does not return to the height of the platform.



MARK SCHEME — 6-MARK QUESTION

Tutor reveal · M = method, A = accuracy, B = independent mark, ECF = error carried forward

Q3

[6 marks]

Level 3 (5–6) a complete, linked description of A→B and B→C using correct store names, including both the elastic potential store and dissipation to the thermal store, with an explanation for the lower return height.

Level 2 (3–4) most transfers described with correct stores, or the dissipation point made but one section of the fall missing.

Level 1 (1–2) isolated correct stores (e.g. “GPE becomes KE”) with no sequence.

Indicative content: A→B (cord slack): gravitational potential store empties, kinetic store fills (mechanically). B→C (cord stretching): kinetic store AND gravitational potential store empty; elastic potential store of the cord fills; at C kinetic energy is zero momentarily and the elastic potential store is greatest. On the way back the elastic potential store empties into kinetic and gravitational potential. Air resistance and internal friction in the cord dissipate energy by heating to the thermal store of the air and cord, so the jumper has less energy and does not return to the platform height.

Max 6 marks

Answer: **Levels-based: see descriptors**

Watch out — and what to remember

IN A NUTSHELL

- $E_e = ½ k e^2$ — squared, so rearrange directly, same as E_k.
- GPE lost = KE gained (no air resistance) links every equation from Part 1.
- $v = \sqrt{(2 g h)}$ is the mass-cancels shortcut for a falling object's landing speed.

COMMON SLIP-UPS

Dropping the limit-of-proportionality condition

$E_e = ½ k e^2$ only holds up to the limit of proportionality — always state this when asked when the equation applies.

Getting stuck on the mass cancelling

In $m g h = ½ m v^2$, both sides have one m — cancel it early rather than trying to substitute a mass you don't need.

Calling dissipated energy “destroyed”

Energy lost to friction and air resistance isn't destroyed — it's transferred by heating to the thermal store of the surroundings.

NEXT Deck 4 — Quick recap & multi-step challenges →